

# Conjugacy classes of $A_n$

Arham Lodha

November 09, 2023

**Problem 1.** Show that  $G$  naturally permutes the  $N$ -orbits, indeed show that  $g(Nx) = N(gx)$ . This gives an action of  $G$  on  $X/N$ . Show the natural  $X \rightarrow X/N$  is a map of  $G$ -sets. Show that for the  $G$  action on  $X/N$ ,  $N$  acts trivially, and so the action of  $G$  on  $X/N$  factors through the quotient  $G \rightarrow G/N$ .

*Proof.* Because  $N \trianglelefteq G$ ,  $\forall g \in G, gN = Ng$  so  $g(N \cdot x) = (gN) \cdot (x) = (Ng) \cdot (x) = N(gx)$ . The natural  $f : X \rightarrow X/N$  is just  $x \rightarrow N \cdot x$  by the previous statement  $f(gx) = N \cdot gx = gN \cdot x = gf(x)$ . Thus  $f$  is a homomorphism of  $G$  sets.

For the  $g$  action on  $X/N$ ,  $N \cdot (N \cdot x) = N \cdot \{n \cdot x : n \in N\} = \{n_1 \cdot (n_2 \cdot x) : n_1, n_2 \in N\} = \{n \cdot x : n \in N\} = N \cdot x$ .

Since  $N$  is in the kernel of the  $G$  action of  $X/N$  and is the kernel of  $G \rightarrow G/N$  by prehomework the action factors through the quotient.  $\square$

**Problem 2.** Now suppose  $G$  acts transitively on  $X$ . Explain why  $N$  need not act transitively on  $X$ , but show that  $G/N$  acts transitively on  $X/N$ . In particular if  $G/N$  is finite, then  $X/N$  is finite and its order divides  $|G/N|$ .

*Proof.* If  $G$  acts transitively on  $X$  it just means when a subgroup of  $G$  acts on  $X$  you get  $X$ . The subgroup need not be normal.

Because  $G$  acts transitively on  $X$ ,  $\forall x, y \in X, \exists g \in G \ni g \cdot x = y$ . Thus  $gN \cdot (N \cdot x) = g \cdot (N \cdot x) = N \cdot (g \cdot x) = N \cdot y$ . Thus  $|G/N \cdot X/N| = 1$ .

By orbit stablizer,  $|G/N| = |X/N| |G/N^{N \cdot x}|$ . Thus  $|X/N|$  is finite and must divide  $|G/N|$ .  $\square$

**Problem 3.** Suppose now  $G$  is finite, and acts transitively on  $X$ , and that  $N$  has prime index,  $p$ , in  $G$ . Show that either  $N$  acts transitively on  $X$ , or there are exactly  $p$   $N$ -orbits in  $X$ , all of the same size. In the second case, we'll say " $X$ -breaks up" (into several equal size  $N$ -orbits)

*Proof.* By previous problem,  $|G/N| = |X/N| |G/N^{N \cdot x}|$  and  $|X/N| \mid p$ . Thus  $|X/N| = 1$  or  $p$ . Thus  $N$  either acts transitively or there are  $p$   $N$ -orbits.  $\square$

**Problem 4.** Same assumptions as in 3) Let  $x$  be in  $X$ . Consider  $N^x$ . Explain why  $N^x = N \cap G^x$  ( $\cap$  means intersection). Explain (e.g. using the midterm, or just prove it here) why either  $G^x = N^x$ , or  $N^x$  has index  $p$  in  $G^x$ . Show that the first case holds iff  $X$  breaks up.

*Proof.* By definition,  $\forall n \in N^x, n \in N \ni n \cdot x = x$ . But  $N \trianglelefteq G$ , then  $n \in G^x$  so  $N^x \in G^x$ . But  $N^x \in N$  so it is all the elements of  $G^x$  which are in  $N$  or  $N^x = N \cap G^x$ .

By the midterm, we know  $|G/N^x| = |G/G^x| |G^x/N^x|$ .  $|G/N^x| = |G/N| |N/N^x| = p |N/N^x|$ . By orbit stablizer,  $|G/G^x| = |X|$ . Thus  $p |N/N^x| = |X| |G^x/N^x|$ . By the previous problem  $X$  either breaks up into  $p$   $N$  orbits or  $N$  acts transitively. By orbit stablizer, if  $X$  breaks up,  $|N/N^x| = |X|/p$ , else  $|N/N^x| = |X|$ .

If  $X$  breaks up:  $|X|/p * p = |X| = |X| |G^x/N^x| \implies |G^x/N^x| = 1 \implies N^x = G^x$ .

If  $N$  acts transitively:  $p |X| = |X| |G^x/N^x| \implies |G^x/N^x| = p$  □

**Problem 5.** Now apply this: Take  $G = S_n, N = A_n, p = 2$ , and let  $X = [x]$ , the  $S_n$ -conjugacy class of some element of  $S_n$ , with  $G$  acting on  $X$  by conjugation. Show that if  $x$  is not in  $A_n$ , then  $A_n$  acts transitively on  $[x]$ . If  $x$  is in  $A_n$ , show that  $[x]$  breaks up iff  $Z(x)$  is contained in  $A_n$ .

*Proof.* By the previous problem we know, if  $A_n^x = S_n^x \iff [x]$  breaks up.  $S_n^x = Z(x)$  by definition. Thus  $A_n^x = S_n^x \cap A_n = Z(x) \cap A_n = Z(x) \implies Z(x) \subseteq A_n$  since  $x \in Z(x)$  then  $x \in A_n$  and  $Z(x) \subseteq A_n \iff [x]$  breaks up. But if  $x \notin A_n \implies Z(x) \not\subseteq A_n \implies [x]$  doesn't break up but by the previous problem then  $A_n$  acts transitively on  $[x]$ . □

**Problem 6.** Write down the " $S_n$ -action equation" for the conjugation action of  $S_n$  on  $A_n$  - i.e. Write  $|A_n|$  as the sum of the orders of the  $S_n$  conjugacy classes in  $A_n$ , for  $n = 3, 4, 5$ . This should be easy (because conjugation in  $S_n$  is easy). Now use this, and the above, to find the class equation for  $A_n$  for  $n = 3, 4, 5$ .

*Proof.* Any permutation in  $S_n$  can be constructed with Young diagram and then conjugation.  $\sum_{i=1}^k n_i = n$  where  $n_i$  describes the size of the cycle and  $n_1 \leq \dots \leq n_k$  then the corresponding permutation would be  $(a_1 \dots a_{n_1})(a_{n_1+1} \dots a_{n_1+n_2}) \dots (a_{\sum_{i=1}^{k-1} n_i+1} \dots a_n)$  where  $a_i$  are distinct and all conjugations of this are just parmuations of  $a_i$  which are permutations of  $n$  elements. The number of conjugations is equal to  $\frac{n!}{(\prod_{i=1}^k n_i) r_i!}$  so the naive permutation of  $n$  elements divided by the permutations which simply cycle a one of the cycles and divided by the permutations which permute the  $r_i$  cycles.

For  $n = 3$ . The possible sums whose corresponding permutations are in  $A_3$  are  $1 + 1 + 1$  or  $3$ . If it is  $1 + 1 + 1$ , the permutation is the identity which is only conjugate to itself. If it is  $3$ , using our formula we se the number of possible conjugations are  $3!/3 = 2$ . Thus  $|A_3| = 1 + 2 = 3!/2$ .

For  $n = 4$ . The possible sums whose corresponding permutations are in  $A_4$  are  $1 + 1 + 1 + 1$ ,  $1 + 3$ ,  $2 + 2$ . If it is  $1 + 1 + 1 + 1$ , the permutation is the identity which is only conjugate to itself. If it is  $1 + 3$ , using our formula we se the number of possible conjugations are  $4!/3 = 4 * 2 = 8$ . If it is  $2 + 2$ , using our formula we have  $4!/(2 * 2 * 2) = 3$  Thus  $|A_4| = 1 + 3 + 8 = 12 = 4!/2$ .

For  $n = 5$ . The possible sums whose corresponding permutations are in  $A_5$  are  $1 + 1 + 1 + 1 + 1$ ,  $1 + 1 + 1 + 3$ ,  $2 + 2 + 1$ , and  $5$ . If it is  $1 + 1 + 1 + 1 + 1$ , the permutation is the identity which is only conjugate to itself. If it is  $1 + 1 + 1 + 3$ , using our formula we se the number of possible conjugations are  $5!/(3 * 2) = 120/6 = 20$ . If it is  $2 + 2 + 1$ , using our formula we have  $5!/(2 * 2 * 2) = 120/8 = 15$ . If it is a  $5$ ,  $5!/5 = 4! = 24$  Thus  $|A_5| = 1 + 15 + 20 + 24 = 60 = 5!/2$ .

Class equations of  $A_n$ .

For  $A_3 \cong C_3$ , since  $C_3$  is in the center of  $D_3 \cong S_3$ . The 2 conjugacy class breaks up  $|A_3| = 1 + 1 + 1$ .

The element  $g$  in the  $S_4$  conjugacy class of size 8 is a 3 cycle.  $|Z(g)| = 3 = |\langle g \rangle|$ . Thus  $\langle g \rangle = Z(g)$  thus  $Z(g) \in A_4$ . Thus 8 breaks up into 4 and 4.  $|A_4| = 1 + 3 + 4 + 4 = 12$ .

The element  $g$  in the conjugacy class of size 20, are all the 3 cycles. The transposition of the 2 elements in the 2 cycle commutes with the 3 cycle. But individual transpositions are not in  $A_5$ . So 20 doesn't break.

The element  $g$  in the conjugacy class of size 24 is a 5 cycle. Cyclic group of  $g$  has order 5 and the centralizer also has order 5. Thus the cyclic group of order 5 must be the centralizer since the cyclic group is a subgroup of the centralizer. Thus 24 breaks up.  $|A_5| = 1 + 12 + 12 + 15 + 20 = 60$ .  $\square$